Technical Report: Bokone Community Network

CMPG 325 – Computer Networks | Project Technical Report

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Client: Bokone Community Network (Vryburg), Client ID CLI-148 | **Industry:** Telecommunications

Assigned address block: 172.30.100.0/23 | **Assigned technical challenge:** EtherChannel (LACP)

Tools: Cisco Packet Tracer, GitHub (CMPG325-Bokone-Community-Network)

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1. Executive Summary

This report documents the design, implementation and testing of a campus network for Bokone Community Network, a telecommunications organisation in Vryburg. The network was designed in a hierarchical model (router, core, distribution, access), segmented into VLANs, addressed from the assigned 172.30.100.0/23 block, and built and tested in Cisco Packet Tracer.

The assigned technical challenge was EtherChannel. Two gigabit links between the core and distribution switches were bundled into one logical Port-channel using the Link Aggregation Control Protocol (LACP). Testing confirmed that the bundle forms correctly on both switches, that routed traffic crosses it, and that connectivity continues with no packet loss when one member link is removed. The network therefore removes the single point of failure that a single uplink would create.

2. Introduction

Technical staff at Bokone depend on the internal network to monitor, manage and support telecommunications services. A failure in the network can delay their work, so reliability is the main design priority. The client also expects logical organisation of users and devices, efficient use of addresses, and room for future growth.

This report explains what was required, how the network was designed, how each device was configured and why, how the network was tested, and what could be improved. Network terms are defined where they are first used.

3. Objectives and Scope

Objectives

1. Design a network that is reliable, segmented by function, efficiently addressed and scalable.
2. Implement the design in Cisco Packet Tracer, with working VLANs, inter-VLAN routing and DHCP.
3. Implement EtherChannel between two switches and verify it, including behaviour when a member link fails.
4. Document the design, configuration, testing and troubleshooting in a GitHub portfolio.

In scope: the Packet Tracer build of one floor (router, core, distribution and access layers, with end devices), and addressing and design provision for a future floor and a future branch office.

Out of scope: physical implementation of the future floor and the branch office (addressing and design only), and a simulated ISP or Internet connection. External connectivity was not a stated client requirement.

4. Client Requirements

Requirement	Design response
Reliable internal connectivity	Hierarchical topology with Layer 3 routing at the core

Requirement	Design response
Logical separation of functions	VLAN and subnet-based segmentation
Communication between VLANs	Layer 3 routing on the core switch (SVIs)
Efficient use of 172.30.100.0/23	/26 subnets sized to the current need, with spare space
Future additional floor	VLAN 50 and subnet 172.30.101.0/26 reserved
Future branch office (CR6)	VLAN 60 and subnet 172.30.101.64/26 reserved
Reliable switch-to-switch link	EtherChannel using LACP
Verified operation	End-to-end, routing, DHCP, and EtherChannel testing

5. Network Design

5.1 Design approach

The network follows a **hierarchical (tree) model** with three layers. The **core** layer routes traffic between VLANs and provides the fast backbone. The **distribution** layer aggregates the access layer and connects it to the core. The **access** layer connects end devices. At the access switch, the end devices form a star because each connects directly to the switch.

This model was chosen because each layer has a clear role, faults are easy to isolate, and new floors or segments can be added at the access or distribution layer without redesigning the core.

5.2 Physical topology

Device	Model	Role
BOKONE-EDGE-RTR01	Cisco ISR4321	Edge router; routed point-to-point link to the core
BOKONE-CORE-SW01	Cisco 3560-24PS (Layer 3)	Inter-VLAN routing, DHCP server, one end of the EtherChannel
BOKONE-DIST-SW01	Cisco 3560-24PS	Other end of the EtherChannel; aggregates the access layer
BOKONE-ACC-SW01	Cisco 3560-24PS	Connects PC1, PC2, PC3 and SRV1

Link	Ports	Type
Router to core	Router Gi0/0/1 to Core Fa0/24	Routed point-to-point, 172.30.101.128/30

Link	Ports	Type
Core to distribution	Core Gi0/1 and Gi0/2 to Dist Gi0/1 and Gi0/2	EtherChannel (Port-channel 1, LACP), 802.1Q trunk
Distribution to access	Dist Fa0/24 to Acc Gi0/1	802.1Q trunk, VLANs 20 and 30
Access to end devices	Acc Fa0/1 and Fa0/2 (PC1, PC2); Fa0/3 and Fa0/4 (PC3, SRV1)	Access ports

5.3 Logical design: VLANs

A **VLAN** (Virtual Local Area Network) divides one switched network into separate broadcast domains. Each VLAN has its own subnet, which improves security, reduces unnecessary traffic and makes the network easier to manage. Devices in different VLANs can communicate only through a Layer 3 device.

VLAN	Name	Purpose	Addressing
10	MGMT	Network management (no end-user devices)	Static
20	STAFF	Staff computers (PC1, PC2)	DHCP
30	NOC_SRV	Network Operations Centre and servers (PC3, SRV1)	Static
50	FUTURE_FLOOR	Reserved for an additional floor	Not configured
60	BRANCH	Reserved for the branch office (CR6)	Not configured

The Network Operations Centre (NOC) is the team that monitors and maintains the network. Its workstations and the servers share VLAN 30 because the network is small and the NOC is the main user of the servers. Because spare address space exists, the two can be separated later if the client grows.

Staff devices use DHCP because they change often and automatic addressing reduces administration. The servers and NOC devices use static addresses so that they are always found at a predictable address.

5.4 IP addressing plan

The assigned block 172.30.100.0/23 is private address space (RFC 1918) and provides 512 addresses. It was divided into /26 subnets (mask 255.255.255.192), each with 64 addresses and 62 usable hosts. This avoids wasting addresses while leaving whole subnets free for growth.

VLAN	Subnet	Usable range	Gateway (SVI)	Status
10 MGMT	172.30.100.0/26	.1 – .62	172.30.100.1	Active
20 STAFF	172.30.100.64/26	.65 – .126	172.30.100.65	Active (DHCP)
30 NOC_SRV	172.30.100.128/26	.129 – .190	172.30.100.129	Active (static)
Spare	172.30.100.192/26	.193 – .254	–	Unallocated

VLAN	Subnet	Usable range	Gateway (SVI)	Status
50 FUTURE_FLOOR	172.30.101.0/26	.1 – .62	–	Reserved
60 BRANCH	172.30.101.64/26	.65 – .126	–	Reserved (CR6)
Router-core link	172.30.101.128/30	.129 – .130	–	Routed link
Spare	172.30.101.192/26	.193 – .254	–	Unallocated

The router-core link uses a /30 subnet (255.255.255.252), which has exactly two usable addresses, so no addresses are wasted on a link that needs only two devices. Router: 172.30.101.129. Core: 172.30.101.130.

Static addresses in VLAN 30: PC3 172.30.100.130 and SRV1 172.30.100.131, both with mask 255.255.255.192 and gateway 172.30.100.129.

5.5 Key design decisions

Decision	Justification
Layer 3 switch with SVIs for inter-VLAN routing	Routes inside the switch, is faster than router-on-a-stick, and needs no extra hardware
Routed point-to-point link to the router	Keeps routing traffic off the VLANs and uses only two addresses
/26 subnets	62 hosts per VLAN suits current needs; whole subnets stay free for growth
DHCP for staff, static for servers	Convenience for users; predictability for servers
EtherChannel with LACP between core and distribution	Adds bandwidth and removes the single point of failure on the key link
VLAN 10 not carried to the access layer	Management is not needed there; allowing only required VLANs on a trunk is good practice

6. Implementation

The following describes what was configured on each device and why. Command excerpts are shown to match the running configuration.

6.1 BOKONE-CORE-SW01 (core switch)

```
hostname BOKONE-CORE-SW01
ip routing

vlan 10
 name MGMT
vlan 20
```

```
 name STAFF
vlan 30
 name NOC_SRV

interface FastEthernet0/24
 no switchport
 ip address 172.30.101.130 255.255.255.252

interface GigabitEthernet0/1
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30
 switchport mode trunk
 channel-group 1 mode active
(GigabitEthernet0/2 is configured identically)

interface Vlan10
 ip address 172.30.100.1 255.255.255.192
interface Vlan20
 ip address 172.30.100.65 255.255.255.192
interface Vlan30
 ip address 172.30.100.129 255.255.255.192

ip route 0.0.0.0 0.0.0.0 172.30.101.129
```

- `ip routing` enables Layer 3 functions. Without it the switch only switches frames and cannot route between VLANs.
- A **Switched Virtual Interface (SVI)** is a virtual Layer 3 interface on a switch, one per VLAN. It carries the VLAN's IP address and acts as its default gateway.
- `no switchport` on Fa0/24 turns it into a **routed port**, which behaves like a router interface and carries its own IP address.
- A **trunk** carries several VLANs over one link. **802.1Q (dot1q)** is the standard that tags each frame with its VLAN identifier. `allowed vlan` limits the trunk to the VLANs that are needed.
- The **default route** (0.0.0.0/0) sends traffic for unknown destinations to the router.
- A **DHCP pool** for VLAN 20 supplies staff computers with an address, mask, default gateway (172.30.100.65) and DNS server (8.8.8.8). The Dynamic Host Configuration Protocol (DHCP) assigns these settings automatically, using four steps: Discover, Offer, Request and Acknowledge.

6.2 BOKONE-EDGE-RTR01 (router)

```
hostname BOKONE-EDGE-RTR01

interface GigabitEthernet0/0/1
 description P2P to BOKONE-CORE-SW01
 ip address 172.30.101.129 255.255.255.252

ip route 172.30.100.0 255.255.254.0 172.30.101.130
```

The router's active interface faces the core. A **static route** is a manually configured instruction that tells a router which next hop to use for a destination network. This route covers the whole 172.30.100.0/23, so the router can return traffic to every VLAN, including the reserved future floor and branch subnets, without further changes.

6.3 BOKONE-DIST-SW01 (distribution switch)

```
vlan 10
 name MGMT
vlan 20
 name STAFF
vlan 30
 name NOC_SRV

interface GigabitEthernet0/1
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30
 switchport mode trunk
 channel-group 1 mode active
(GigabitEthernet0/2 is configured identically)

interface FastEthernet0/24
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 20,30
 switchport mode trunk
```

This is a Layer 2 switch, so it holds the same VLANs as the core but does not route. The EtherChannel member ports match the core's settings exactly, which is a requirement for the bundle to form. Fa0/24 trunks to the access switch and carries only VLANs 20 and 30.

6.4 BOKONE-ACC-SW01 (access switch)

```
vlan 20
 name STAFF
vlan 30
 name NOC_SRV

interface GigabitEthernet0/1
 switchport mode trunk

interface FastEthernet0/1
 switchport mode access
 switchport access vlan 20
(FastEthernet0/2 is identical, for PC2)

interface FastEthernet0/3
 switchport mode access
 switchport access vlan 30
(FastEthernet0/4 is identical, for SRV1)
```

An **access port** belongs to a single VLAN and connects an end device. The port assignment decides which VLAN a device joins, so end devices need no special configuration.

7. Assigned Technical Challenge: EtherChannel

7.1 Concept

If two switches are joined by two parallel cables, the **Spanning Tree Protocol (STP)**, which prevents switching loops, blocks one of them. The client then pays for two links but uses only one, and the blocked link is idle until a failure occurs.

EtherChannel bundles several physical links into one logical link, called a Port-channel. STP treats the bundle as a single link, so none of its members is blocked. The benefits are:

- **Higher bandwidth:** traffic is shared across the member links by flow, so many conversations are spread across both.
- **Redundancy:** if one member fails, traffic continues on the others without reconfiguration.
- **Simpler management:** the bundle is configured and monitored as one interface.

7.2 LACP

The **Link Aggregation Control Protocol (LACP)**, defined in IEEE 802.3ad, lets the two switches negotiate the bundle. Each side exchanges LACP messages and the ports join the bundle only if their settings agree. This prevents misconfiguration, whereas a static bundle (`mode on`) forms without any check and can cause loops if the settings differ.

In **active** mode a port starts the negotiation. In **passive** mode it only replies. A bundle forms if at least one side is active. Both switches here use active mode. LACP is an open standard, so it also works between different vendors.

7.3 Configuration

On both CORE-SW01 and DIST-SW01, interfaces Gi0/1 and Gi0/2 were configured with the same trunk settings and placed in channel-group 1 using `mode active`. All member ports must match in speed, duplex, trunk mode and allowed VLANs, otherwise the bundle does not form.

7.4 Verification

The command `show etherchannel summary` was run on both switches:

```
Group  Port-channel  Protocol  Ports
1      Po1(SU)       LACP      Gig0/1(P)  Gig0/2(P)
```

- **Po1:** the logical Port-channel interface.
- **S:** Layer 2 bundle. **U:** in use.
- **P:** the member port is bundled in the Port-channel.
- **LACP:** the negotiation protocol in use.

The same result on both switches confirms that they negotiated successfully.

8. Testing and Verification

ID	Test	Method	Result
T1	Normal connectivity and inter-VLAN routing	Ping from PC1 (VLAN 20) to 172.30.100.129 (VLAN 30 gateway)	4 of 4 replies, 0% loss
T2	Link failure	One of the two core-to-distribution cables removed while the network ran; same ping repeated	4 of 4 replies, 0% loss
T3	DHCP for VLAN 20	`ipconfig /release` and `/renew` on PC2; PC1 configured by DHCP; `show ip dhcp binding` on the core	PC1 received 172.30.100.71 and PC2 172.30.100.72, mask 255.255.255.192, gateway 172.30.100.65, DNS 8.8.8.8
T4	Routing path	`tracert 172.30.100.129` from PC1	Completed in one hop, confirming the core performs the routing
T5	Static addressing in VLAN 30	Ping from PC3 and from SRV1 to 172.30.100.129	4 of 4 replies from each, 0% loss
T6	EtherChannel formation	`show etherchannel summary` on both switches	Po1(SU) with Gig0/1(P) and Gig0/2(P), protocol LACP
T7	Router to core	Ping from the router to 172.30.100.65	4 of 5 replies; the first timed out while the router resolved the next hop's address (ARP), then 0 further loss
T8	Core-to-router path from the user side	Ping from PC1 to the router address 172.30.101.129	Successful

An ARP timeout on the first packet is normal behaviour. The device must learn the next hop's hardware address before it can send the first packet.

9. Troubleshooting and Issues Resolved

Issue	Cause	Resolution
The router had no route back to the VLAN subnets, and its default route pointed at the core while	The initial configuration used a default route in both directions, which	The router's default route was removed and replaced with a specific static route, `ip route 172.30.100.0 255.255.254.0 172.30.101.130`. After the change the router

Issue	Cause	Resolution
the core's default route pointed at the router	would loop any traffic for unknown destinations	reached 172.30.100.65 and PC1 reached the router
EtherChannel failure risk from mismatched member ports	A bundle does not form unless both ends match	Identical settings were applied to Gi0/1 and Gi0/2 on both switches, and the result was verified with `show etherchannel summary`

10. Scalability and Future Growth

- **Additional floor:** VLAN 50 and subnet 172.30.101.0/26 are reserved. To add the floor, create VLAN 50, add an SVI with gateway 172.30.101.1, and add VLAN 50 to the trunks. No readdressing is required.
- **Branch office (CR6):** VLAN 60 and subnet 172.30.101.64/26 are reserved for the same approach.
- **Router:** the static route already covers the whole /23, so the router needs no change for either extension.
- **Spare space:** two further /26 subnets (172.30.100.192/26 and 172.30.101.192/26) remain unallocated.
- **EtherChannel:** further member links can be added to the bundle if bandwidth demand grows.

11. Limitations and Recommendations

Limitation	Recommendation
One edge router is a single point of failure	Add a second router with a first-hop redundancy protocol such as HSRP
The WAN or ISP connection is not simulated, because Internet access was not a stated requirement	Add an ISP link with a default route and NAT when external access is required
No traffic filtering between VLANs	Add access control lists (ACLs) between VLANs
Access ports are open to any device	Enable port security on the access ports
Only one distribution switch and one access switch	Add redundant uplinks and switches as the network grows

12. Conclusion

The network meets the Bokone Community Network requirements. It is segmented into VLANs, efficiently addressed from the 172.30.100.0/23 block, routed by a Layer 3 core switch, and automated with DHCP for staff. It is ready to grow, with VLAN 50 and VLAN 60 reserved and a router route that already covers the full address block.

The assigned EtherChannel challenge was implemented with LACP between the core and distribution switches and verified on both switches. Testing showed that traffic crosses the bundle and that connectivity continued with no packet loss when one member link was removed. This gives the client a resilient, higher-bandwidth link between the core and distribution layers and removes the single point of failure that a single cable would have created.

References

Cisco Systems, Inc. (2023) *Understand EtherChannel load balance and redundancy on Catalyst switches.* Available at: https://www.cisco.com/c/en/us/support/docs/lan-switching/etherchannel/12023-4.html (Accessed: 25 August 2026).

Cisco Systems, Inc. (2024) *Configure Inter-VLAN Routing with Catalyst Switches.* Cisco Systems, Inc. (Accessed: 25 August 2026).

Internet Engineering Task Force (1996) *RFC 1918: Address Allocation for Private Internets.* (Accessed: 26 August 2026).

Appendix A: Portfolio Evidence

The GitHub repository **CMPG325-Bokone-Community-Network** contains the client requirements, topology diagrams, IP addressing plan, the Packet Tracer file (Bokone_community_network.pkt), device configurations, and the screenshots for tests T1 to T8.

GitHub Link [CMPG325-Bokone-Community-Network]

Appendix B: Verification Command Reference

Command	Purpose
`show vlan brief`	Lists VLANs and their ports
`show interfaces trunk`	Shows trunk ports and the VLANs they carry
`show ip interface brief`	Shows interface addresses and status, including SVIs
`show ip route`	Shows the routing table
`show ip dhcp binding`	Lists the addresses issued by DHCP
`show etherchannel summary`	Shows bundle status, protocol and member port flags
`tracert <address>`	Shows the path and hop count from a PC
`ping <address>`	Tests reachability